

Generation Interconnection System Impact Study Report

For Queue Project Q16-0616

Nueva Guinea 3405 138 kV

March 15, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0616. The Interconnection Customer proposes a 97.2 MW solar photovoltaic facility interconnecting at the Nueva Guinea 3405 138 kV substation in the Pennsylvania area. The request was received on May 08, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 327 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,375,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The MOROVIS 8801 - BULLMOOSE SUBSTATION 138 kV is overloaded to 239.9% of its 248 MVA emergency rating for the loss of the WEST GALLATIN - WALNUT SUBSTATION 345 kV. The project contributes 7.1% DFAX (6.9 MW) to this overload.

The NEWCOMERSTOWN - MOROVIS 8801 138 kV is overloaded to 211.7% of its 289 MVA emergency rating for the loss of the W GALLATIN - WALNUT 345 kV. The project contributes 7.1% DFAX (6.9 MW) to this overload.

For the loss of WEST GALLATIN - WALNUT 345 kV, the NEWCOMERSTOWN - VALLEY HOME TAP 2 SUBSTATION 138 kV line loads to 192.8% of its 341 MVA rating. Project DFAX: 7.1%.

For the loss of WEST GALLATIN - WALNUT 345 kV, the EAST ARLINGTON - VALLEY HOME TAP 2 138 kV loads to 186.8% of its 362 MVA rating. Project DFAX: 7.1%.

The WESCOSVILLE - DORNYVILLE 138 kV is overloaded to 185.3% of its 213 MVA emergency rating for the loss of the WEST GALLATIN - WALNUT SUBSTATION 345 kV. The project contributes 6.3% DFAX (6.1 MW) to this overload.

E ARLINGTON - WESCOSVILLE 138 kV line: 173.5% loading (218 MVA) following the outage of the WEST GALLATIN - WALNUT 345 kV; DFAX = 6.3%.

EDWARDSPORT STATION - DORNYVILLE 138 kV: 166.3% loading (251 MVA) following the outage of the W GALLATIN - WALNUT 345 kV L; DFAX = 6.3%.

MOUNT OLIVE - EDWARDSPORT STA 138 kV: 161.5% loading (272 MVA) following the outage of the W GALLATIN - WALNUT 345 kV; DFAX = 6.3%.

VANDALIA - MOUNT OLIVE 138 kV Line: 133.6% loading (269 MVA) following the outage of the W GALLATIN - WALNUT 345 kV L; DFAX = 5.0%.

For the loss of ESQUON - KOOSKIA SUBSTATION 138 kV L, the RAILHEAD - S MTN 138 kV loads to 124.7% of its 249 MVA rating. Project DFAX: 6.8%.

POLK ST - VAN WAGNER SUBSTATION 138 kV: 123.8% loading (393 MVA) following the outage of the WALNUT - WEST GALLATIN 345 kV; DFAX = 5.8%.

The WHITE SWAN - S MTN 138 kV is overloaded to 122.6% of its 260 MVA emergency rating for the loss of the ESQUON - KOOSKIA 138 kV. The project contributes 6.8% DFAX (6.6 MW) to this overload.

The RAILHEAD - POLK ST 138 kV is overloaded to 122.4% of its 408 MVA emergency rating for the loss of the WEST GALLATIN - WALNUT 345 kV. The project contributes 5.8% DFAX (5.6 MW) to this overload.

The CLEAR SPRINGS 500/345kV XFMR is overloaded to 113.6% of its 1,528 MVA emergency rating for the loss of the HILLS CHAPEL - BURNS 345 kV. The project contributes 10.6% DFAX (10.3 MW) to this overload.

The WALNUT - CLEAR SPRINGS 345 kV is overloaded to 109.8% of its 1,521 MVA emergency rating for the loss of the MINNEAPOLIS CITY OF - WALLENPAUPACK 500 kV. The project contributes 8.1% DFAX (7.8 MW) to this overload.

For the loss of NUEVA GUINEA 3405 - WESTSIDE 138 kV line, the CLEAR CROSSING - PULPGEN 138 kV loads to 104.9% of its 303 MVA rating. Project DFAX: 6.8%.

For the loss of NUEVA GUINEA 3405 - WESTSIDE 138 kV, the WINDSTONE - CLEAR CROSSING 138 kV loads to 104.3% of its 313 MVA rating. Project DFAX: 6.8%.

WIGWAM - UPSCO 345 kV: 104.3% loading (1,213 MVA) following the outage of the W GALLATIN - WALNUT 345 kV; DFAX = 8.9%.

WALDEN NORTH GENEATION STATION - WINDSTONE 138 kV: 103.4% loading (330 MVA) following the outage of the NUEVA GUINEA 3405 - WESTSIDE 138 kV; DFAX = 6.8%.

For the loss of WALUT - W GALLATIN 345 kV, the UPSCO - WHEELABRATOR FRACKVILLE ENERGY 345 kV loads to 102.7% of its 1,225 MVA rating. Project DFAX: 9.2%.

WALDEN NORTH GENERATION STA SUBSTATION - LOS ANGELES HTS 138 kV: 102.6% loading (348 MVA) following the outage of the NUEVA GUINEA 3405 - WESTSIDE 138 kV; DFAX = 6.8%.

VAN WANER - LOS ANGELES HTS 138 kV: 102.2% loading (358 MVA) following the outage of the NUEVA GUINEA 3405 - WESTSIDE SUBSTATION 138 kV L; DFAX = 6.8%.

The WEST GALLATIN - WALNUT 345 kV is overloaded to 100.6% of its 1,770 MVA emergency rating for the loss of the WALLENPAUPACK 500-345 kV Transformer Bank. The project contributes 9.2% DFAX (8.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BULLMOOSE - MOROVIS 8801 138 kV Line	Rebuild line with higher-capacity conductor	\$1,777,000	100% (\$1,777,000)
NEWCOMERSTOWN - MOROVIS 8801 138 kV	Rebuild line with higher-capacity conductor	\$3,027,000	100% (\$3,027,000)
NEWCOMERSTOWN - VALLEY HOME TAP 2 138 kV	Rebuild line with higher-capacity conductor	\$3,625,000	100% (\$3,625,000)
E ARLINGTON - VLY HOME TAP 2 SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$2,386,000	100% (\$2,386,000)
DORNYVILLE - WESCOSVILLE 138 kV L	Rebuild line with higher-capacity conductor	\$2,738,000	100% (\$2,738,000)
E ARLINGTON - WESCOSVILLE 138 kV line	Rebuild line with higher-capacity conductor	\$1,516,000	100% (\$1,516,000)
EDWARDSPORT STATION - DORNYVILLE 138 kV	Rebuild line with higher-capacity conductor	\$8,225,000	100% (\$8,225,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MOUNT OLIVE - EDWARDSPOINT STA 138 kV	Rebuild line with higher-capacity conductor	\$1,194,000	100% (\$1,194,000)
VANDALIA - MOUNT OLIVE 138 kV Line	Rebuild line with higher-capacity conductor	\$5,827,000	100% (\$5,827,000)
RAILHEAD - S MTN 138 kV	Rebuild line with higher-capacity conductor	\$5,904,000	41% (\$2,398,000)
POLK ST - VAN WAGNER SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$12,611,000	24% (\$2,979,000)
WHITE SWAN - S MTN 138 kV	Rebuild line with higher-capacity conductor	\$13,127,000	41% (\$5,331,000)
RAILHEAD - POLK ST 138 kV	Rebuild line with higher-capacity conductor	\$4,531,000	24% (\$1,070,000)
CLEAR SPRINGS 500/345kV XFMR	Replace transformer with higher-rated unit	\$18,756,000	2% (\$341,000)
WALNUT - CLEAR SPRINGS 345 kV	Reconductor line	\$14,810,000	2% (\$261,000)
CLEAR CROSSING - PULPGEN 138 kV	Reconductor line	\$3,643,000	59% (\$2,152,000)
WINDSTONE - CLEAR CROSSING 138 kV	Reconductor line	\$1,089,000	59% (\$643,000)
WIGWAM - UPSCO 345 kV line	Reconductor line	\$27,441,000	2% (\$489,000)
WALDEN NORTH GENEATION STATION - WINDSTONE 138 kV	Reconductor line	\$3,452,000	59% (\$2,039,000)
UPSCO - WHEELABRATOR FRACKVILLE ENERGY 345 kV	Reconductor line	\$8,472,000	2% (\$196,000)
WALDEN NORTH GENERATION STA SUBSTATION - LOS ANGELES HTS 138 kV	Reconductor line	\$1,328,000	59% (\$784,000)
VAN WAGNER - LOS ANGELES HTS 138 kV	Reconductor line	\$1,864,000	59% (\$1,101,000)
WEST GALLATIN - WALNUT 345 kV	Reconductor line	\$16,871,000	2% (\$276,000)
	Total allocated to project		\$50,375,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.